

THE CONSTITUTIONAL SWARM

From Endogenous Coalitions to Endogenous Institutions

A didactic teaching note on how isolated AI agents discover other minds, create a communication space, propose rules, choose a social compact, and organize under the institution they select.

For students moving from traditional financial models toward adaptive, self-organizing agentic systems.

PAPER OVERVIEW

The notebook begins as a controlled M&A search and ends as an experiment in artificial institutional formation. Eighteen specialized agents initially behave as if they are alone. Private search produces local evidence; Claude Sonnet 5 evaluates the semantic value of selected discoveries; high-value findings leave epistemic pheromones; encounters with those traces reveal that other intelligent searchers exist. A dormant bulletin board then becomes socially meaningful. Agents use it to confront a second-order problem: before sharing deeply, how should collaboration work? Competing constitutions are generated, the population chooses among them through an inspectable voting mechanism, and the adopted social compact governs subsequent sharing and coalition formation. The resulting graph and collective hypothesis are therefore products not only of the acquisition problem but of an institution partly generated inside the system.



CENTRAL IDEA

**The problem does not merely select the answer.
It begins to design the organization that produces it.**

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HOW TO READ THIS PAPER

This image-free edition follows the ten code units of the Colab notebook, but it is intentionally not a code commentary. Each unit is interpreted as a step in a social process: substrate, isolated search, semantic judgment, environmental traces, discovery of plurality, constitutional creativity, collective choice, governed coalition formation, collective hypothesis, and institutional inspection. The full illustrated edition contains a corresponding visual for each unit; this GitHub version is designed to compile as a self-contained text document without external image files.

Why This Notebook Matters

Traditional financial models are normally designed as bounded analytical objects. Inputs enter, a specified transformation occurs, and outputs emerge. Even sophisticated machine-learning systems largely preserve this architecture: the model may learn parameters, but the organization surrounding the model remains fixed. Agentic systems change the problem because intelligence can be distributed across multiple components with different capabilities, memories, tools, and objectives. Yet many multi-agent systems still hide a conventional assumption: the organization is designed in advance. Roles are assigned, communication channels are predetermined, and the rules of cooperation are supplied by the programmer.

The Constitutional Swarm asks what happens when we relax that assumption. Its central question is not simply whether several AI agents can jointly find a better acquisition target. The deeper question is whether a population of agents can discover that other minds exist, recognize the value of collaboration, create a public communication surface, invent candidate rules for cooperation, select among those rules, and then allow the chosen institution to shape subsequent collective behavior. The notebook thus turns a familiar financial-practitioner problem into a laboratory for studying endogenous organization.

This is a natural next step for students who have already learned conventional prediction, optimization, and model orchestration. In a traditional model, the object of design is the mapping from data to decision. In a conventional agent workflow, the object of design expands to include roles and handoffs. In the Constitutional Swarm, the object of study expands again: some of the architecture itself becomes a variable. The problem begins to influence not only the answer but the social machinery through which the answer is produced.

THE CENTRAL PROPOSITION

The notebook explores a stronger form of endogeneity: the problem helps generate discoveries; discoveries help generate relationships; relationships create a need for rules; agents generate and choose those rules; and the selected rules influence the organization that ultimately produces the collective answer.

The experiment is deliberately bounded. It does not claim that an artificial society arises from nothing. Identity, compute, documents, model access, persistence, voting infrastructure, and basic communication primitives are supplied by the designer. This is a virtue rather than a weakness. Scientific clarity requires us to distinguish the substrate from what emerges inside it. The relevant research question is therefore: *which organizational decisions can be relocated from the programmer into the adaptive system while remaining observable and governable?*

The General Framework: From the Bulletin Board to the Constitution

The companion framework in *The Bulletin Board: From Artificial Agents to Artificial Societies* begins with an even more general thought experiment: perhaps the minimal institutional seed of a complex artificial organization is simply a place to communicate. Unrelated intelligent agents need not begin as a team. If they can make needs, discoveries, and capabilities publicly visible, latent complementarities can become relationships. Language then becomes an organizational technology: an agent can explain what it knows, identify what it does not know, and ask whether another kind of intelligence may be useful. [2, 1]

The Constitutional Swarm instantiates a more controlled version of that idea. Instead of unrelated tasks, all agents inhabit one synthetic acquisition universe. This common problem gives us a benchmark against which to study institutional formation. Yet the agents are not told that they constitute a team. They first search privately. Only after encountering traces left by other searchers do they infer plurality. Communication is therefore not the starting point of the society; it becomes meaningful after the agents discover that a society might exist.

The framework distinguishes communication infrastructure from organizational architecture. A bulletin board is not an orchestrator. It resembles a public square, an issue tracker, or a trading venue: it makes interaction possible without determining which interaction must occur. This distinction is essential for self-organization. If a central planner assigns every partner and every task, the resulting network may be dynamic, but it is not meaningfully emergent. The notebook instead lets the problem create reasons for connection.

Once agents can communicate, however, a governance problem appears immediately. Should every discovery be shared? Who may join a coalition? What counts as verified evidence? How should credit be allocated? How should groups merge or split? What prevents premature consensus? The possibility of collaboration therefore creates the need for a constitution. This is the conceptual bridge from the general framework to the notebook: *public addressability can produce relationships, but durable collaboration requires rules*. The experiment asks whether some of those rules can themselves become endogenous.

This progression connects to established work on distributed problem solving, contract-based coordination, swarm intelligence, and institutional self-governance. Contract Net showed how distributed nodes can allocate tasks through structured negotiation; swarm research demonstrated how local signals can generate global patterns; Ostrom's work emphasized that communities can create rules governing shared activity rather than relying exclusively on centralized command. [7, 8, 9, 6] The notebook translates these broad ideas into a small, inspectable LLM-era experiment for finance.

Code Unit 1: Runtime, Claude Sonnet 5, and the Minimal Constitutional Substrate

The first code unit establishes the environment in which the constitutional swarm can exist. This is intentionally more than routine setup. In an experiment about endogenous institutions, we must be explicit about what is not endogenous. Every artificial society begins inside a substrate that somebody designed. Here that substrate includes the Python runtime, the Anthropic interface, a model identifier, a finite API-call budget, deterministic random seeds, and utility functions for converting model responses into structured JSON. These elements are analogous to the physical and legal infrastructure within which a human institution operates: they create possibility without specifying the substantive organization that will later emerge.

Claude Sonnet 5 is placed behind one configuration variable. This makes the model a replaceable reasoning component rather than the architecture itself. The distinction is important. The notebook is not demonstrating that a frontier model can solve an acquisition problem by itself. It is demonstrating how model judgment can be embedded inside a larger system whose social structure changes over time. The API-call counter also makes scarcity visible. Intelligence is not treated as free; the experiment has a computational budget.

The code additionally creates the reusable LLM helper that requests structured output. This matters because later stages depend on machine-readable judgments: relevance probabilities, novelty, missing expertise, pheromone text, and constitutional proposals. The helper therefore acts as a narrow institutional interface between probabilistic language-model reasoning and deterministic system logic.

Pedagogically, this unit establishes the central discipline of the notebook: endogeneity is always relative to a substrate. We will allow discoveries, social awareness, constitutions, and coalition structure to emerge, but we will not pretend that the computational universe itself emerged from nothing. Keeping that boundary explicit is essential for both scientific interpretation and governance.

The first design lesson is methodological. A self-organizing system is never literally structureless. It exists inside a substrate: a runtime, an identity scheme, a model interface, resource limits, persistence mechanisms, and permissions. The useful question is therefore not whether everything is endogenous, but which decisions are moved from the designer into the adaptive system. This distinction prevents a common error in discussions of autonomy, where a system is described as self-organizing even though its important social relationships were silently fixed in advance.

For financial practitioners, this is analogous to the difference between a market and the legal infrastructure that makes a market possible. A trading venue may permit decentralized price discovery, but the existence of accounts, settlement, property rights, and admissible instruments remains institutional infrastructure. In the notebook, Claude is similarly not the society. It is one reasoning service available inside a larger institutional experiment. That separation between model capability and system architecture is one of the most important habits students should carry into real financial deployments.

Code Unit 2: Build the Synthetic Acquisition Universe and the Isolated Agents

The second unit creates the world the agents will inhabit. The acquisition universe is synthetic because the purpose is architectural rather than empirical: we want a controlled environment in which the correct target is knowable to the experiment designer but not directly available to any single agent. The hidden company, C0173, is attractive only when multiple fragments of evidence are combined. Financial performance, leverage, desalination technology, intellectual property, ownership structure, and Gulf strategic relevance are deliberately distributed across records. Decoys contain partial matches so that superficial keyword search is insufficient.

This design creates a meaningful reason for heterogeneous expertise. Agents are assigned specialties such as finance, engineering, patents, Gulf strategy, ownership, and synthesis. They also receive bounded search budgets and private state. Most importantly, they begin with a false but operationally useful assumption: each behaves as though it is alone. There is no initial team, no directory of peers, and no orchestrator announcing that complementary expertise

is available.

The code then performs private search. Each specialty interprets the same universe through a different lens, producing different candidate documents and evidence opportunities. This is the first source of structural diversity. The agents are not merely copies of one another running in parallel; their specialties create different local views of the problem.

The conceptual point is that social organization should not be smuggled into the experiment at initialization. If the agents begin as a predefined team, later collaboration proves very little about institutional emergence. By starting with isolated searchers, NB04 creates a genuine transition from individual cognition to social cognition. The problem exists first. Organization must be discovered later. This allows us to observe the moment when independent intelligence becomes aware that additional intelligence may exist and may be useful.

The acquisition case is intentionally small enough to understand but rich enough to require synthesis. C0173 is not identified by a single spectacular clue. Its attractiveness is distributed across financial, technical, strategic, intellectual-property, and ownership evidence. Several decoys satisfy some of the mandate and fail on others. This creates the exact condition under which specialization matters: no local perspective is sufficient.

The eighteen agents are therefore better understood as bounded viewpoints than as eighteen copies of the same chatbot. Finance agents privilege earnings, leverage, and cash conversion. Engineering agents notice membrane performance and operating conditions. Patent agents focus on claims, remaining life, proprietary know-how, and design-around risk. Gulf-strategy agents interpret certification and distribution evidence. Ownership agents look for succession, advisers, liquidity, and willingness to transact. Synthesis agents search more broadly. The resulting diversity is the raw material from which social organization can later emerge.

Code Unit 3: Claude Sonnet 5 as a Judge of Discovery Value

Private search produces candidate evidence, but candidate evidence is not the same thing as a meaningful discovery. This unit introduces the first substantive role for Claude Sonnet 5: semantic valuation. The model examines a local finding in the context of the acquisition mandate and returns a structured judgment about its probability of relevance, discovery value, novelty, why it matters, what expertise is still missing, and what compact trace should be left behind.

This is deliberately different from asking the LLM to “solve the deal.” The model is acting as a local epistemic judge. Its task is bounded: interpret the significance of a discovery made by an agent. This placement makes the architecture easier to inspect because the model’s contribution can be compared with the underlying evidence and with later system behavior.

The distinction between relevance and value is pedagogically important. A document can be relevant without being decisive. A clue about a membrane patent may strongly relate to the mandate, but its value depends on whether it changes confidence in technological defensibility. Novelty matters because repeating an already-known fact should not trigger the same organizational response as revealing a new uncertainty. Missing expertise matters because it transforms a discovery into a potential demand for collaboration.

The output therefore has both epistemic and organizational consequences. Claude’s judgment does not merely update a score. It can identify the boundary of the current agent’s competence and encode a signal that another kind of intelligence would be useful.

This is the first point at which language-model reasoning begins to shape architecture. The LLM is not routing agents through a predefined workflow. Instead, by interpreting what a discovery means and what remains unresolved, it helps create the conditions under which a future relationship may become valuable. Meaning becomes a precursor to organization.

This is the notebook’s first carefully bounded use of generative intelligence. The language model is not asked to solve the acquisition problem from scratch. Deterministic search first compresses the information set, and only then is Claude asked to make a semantic judgment about a limited number of candidate discoveries. This is a practical architecture for finance: inexpensive, auditable computation narrows the field; expensive probabilistic reasoning is reserved for ambiguity.

The output is deliberately structured. Relevance asks whether the evidence materially advances the mandate. Discovery value asks how consequential the clue may be. Novelty distinguishes repetition from information that changes the search. Missing expertise converts uncertainty into a potential demand for another kind of intelligence. The pheromone field compresses the insight into a clue that another specialist could notice. In this sense, the LLM is performing not merely classification but epistemic triage: it judges what is worth carrying forward into the social layer of the system.

Code Unit 4: Epistemic Pheromones: Leaving Traces Without Sending Messages

The fourth unit converts high-value discoveries into epistemic pheromones. The metaphor is intentionally biological. In many decentralized systems, coordination begins not through direct negotiation but through persistent traces left in a shared environment. An ant does not need a global map or a central dispatcher to influence another ant. A local trace changes the probability of another actor's future behavior.

The same principle is used here, but the trace carries epistemic content. When Claude Sonnet 5 judges a discovery sufficiently important, the discovering agent leaves a compact signal in the environment. The trace is not initially a social message. It does not assume that the agent knows who will read it, whether anyone else exists, or whether collaboration is possible. It simply externalizes evidence that some intelligent process found something noteworthy.

This distinction protects the logic of the experiment. If agents immediately addressed one another by name, social awareness would already have been supplied by the programmer. Pheromones allow information to persist before society exists. They create the possibility that one agent can encounter the residue of another agent's cognition and infer plurality from the environment itself.

The unit also illustrates an important design principle for decentralized systems: communication can be indirect before it becomes direct. Shared state can precede shared identity. A trace can precede a relationship.

From a governance perspective, pheromones are also useful because they are inspectable. We can see which discovery generated the trace, what value threshold was crossed, and which semantic summary was exposed. This makes the transition from private reasoning to environmental signaling observable. The system has not yet formed coalitions or constitutions. It has merely created evidence that intelligence leaves footprints—and those footprints can become the first bridge between isolated minds.

Pheromones are the bridge between cognition and sociality. A high-value, sufficiently novel discovery leaves an environmental trace, but the trace is weaker than a direct message. It does not initially reveal the identity of the source agent, and it does not compel a response. This is important because the experiment is trying to show how coordination can begin before a society exists.

The strength of the trace combines relevance and novelty. That construction has a useful interpretation: a clue deserves social attention not simply because it looks related to the mandate, but because it adds something that may change what the group should investigate next. The pheromone therefore externalizes local judgment. A private discovery becomes a public possibility without yet becoming an explicit collaboration request. This is a minimal mechanism of stigmergy: agents coordinate indirectly by modifying a shared informational environment.

Code Unit 5: Discovering Other Minds and Activating the Bulletin Board

This unit contains the notebook's first explicitly social transition. An agent encounters a pheromone that it did not create and infers that another intelligent process must have been active in the environment. The inference is simple, but conceptually it is profound. The agent's model of the world changes from "I am searching alone" to "other minds exist here." That is the birth of social awareness.

The code records this transition in agent state. Once a sufficient number of agents become aware of plurality, the dormant bulletin-board primitive becomes socially active. The board has existed as infrastructure, but until now it had no practical meaning. A communication channel is useful only when an agent has reason to believe that someone else might receive the message.

This is why the bulletin board should not be interpreted as a central orchestrator. It does not decide who should collaborate, allocate tasks, or select the target. It is simply an addressable public surface. Agents can post discoveries, uncertainties, and needs. Other agents can read them. The board transforms unknown potential collaborators into reachable potential collaborators.

The sequence is therefore important: private search comes first; traces appear second; plurality is inferred third; direct public communication becomes rational fourth. The notebook does not begin with a team chat. It tries to model the conditions under which a team chat becomes useful.

Pedagogically, this unit marks the shift from stigmergic coordination to linguistic coordination. Pheromones say, in effect, "something happened here." Bulletin-board messages can say, "this is what I found, this is what I do not understand, and this is the kind of help that might matter." Language therefore expands the organizational bandwidth of the swarm. It allows agents not merely to react to traces but to articulate competence gaps and make

collaboration opportunities legible.

The encounter with another agent's trace is the notebook's ontological turning point. Before the encounter, an agent's world contains documents, companies, and its own search process. After the encounter, the world contains another mind. The inference another intelligent searcher has been here" is small in code but profound conceptually, because it creates the precondition for social organization.

The bulletin board is dormant infrastructure until enough agents infer plurality. Once at least three agents know that others exist, the board becomes socially activated. This avoids treating communication as meaningful merely because an API or data structure exists. A public square becomes an institution only when actors understand that other actors may be present and responsive. The first posts are therefore constitutional calls rather than task assignments: before deep sharing begins, the agents recognize that they need rules. Communication does not solve the governance problem; it reveals it.

Code Unit 6: Claude Sonnet 5 Generates Competing Constitutions

Once agents can communicate, a new problem appears: communication alone does not determine how collaboration should work. Who may join a coalition? What evidence must be shared? How should claims be verified? How are rewards allocated? What prevents herding? When should a coalition split, merge, or dissolve? These are institutional questions rather than search questions.

This unit gives Claude Sonnet 5 its second major role: constitutional creativity. Several agents representing different specialties act as proposers. Each asks the model to design a social compact suited to the emerging environment. The requested constitution contains fields for a guiding principle, sharing threshold, joining rule, verification rule, reward rule, merge/split logic, and anti-herding protection.

The important point is not that the LLM writes elegant policy text. It is that institutional alternatives become objects inside the system. Instead of one collaboration protocol being silently embedded in code, multiple possible protocols are made explicit and contestable. The agents can therefore reason not only about the acquisition target but about the machinery of collaboration itself.

Different proposers may naturally emphasize different values. A finance-oriented proposer may care about incentives and contribution. A patent-oriented proposer may emphasize verification and provenance. A synthesis agent may prefer broader information sharing. The resulting constitutions expose trade-offs that would otherwise remain hidden inside architecture.

This is the notebook's central move from endogenous topology to endogenous rules. In NB03, agents could generate the collaboration graph while operating under rules supplied by the designer. Here the system begins to generate candidate rules governing the graph.

The experiment still has a constitutional substrate—the fields that a proposal must address are specified ex ante—but the substantive answers are produced through model judgment. This bounded openness is intentional: it creates room for institutional invention without making the experiment impossible to inspect.

The constitutional proposals are where language-model creativity moves from task interpretation to institutional design. Three proposers with different specialties are asked to formulate compact social protocols. They must specify sharing thresholds, joining rules, verification, rewards, merger and separation, and a safeguard against herding. The prompt explicitly forbids assuming a central coordinator.

This matters because institutional design is not reducible to choosing a parameter. Different constitutions embody different theories of cooperation. One may privilege openness; another may emphasize verification or property-like control over discoveries; another may protect autonomy and dissent. Claude is useful here because the design space is semantic and combinatorial. Yet the model is not granted sovereign authority. It proposes possibilities. The society will choose among them in the next stage. The separation between constitutional creativity and constitutional legitimacy is deliberate.

Code Unit 7: Collective Choice: Agents Vote on the Social Compact

Generating constitutions is not enough. A society requires some mechanism for selecting which rules will actually govern behavior. This unit therefore introduces collective choice. The candidate constitutions created by Claude Sonnet 5 are evaluated by the agents, and a winning social compact is selected through a transparent voting procedure.

The voting logic is intentionally deterministic. This is a methodological choice rather than a limitation. The LLM has already supplied open-ended institutional creativity; using another model call for every vote would increase cost and make the choice mechanism less inspectable. Instead, agents evaluate proposals according to local incentives and

preferences that can be examined directly in code. The result is a useful separation between institutional generation and institutional selection.

This separation mirrors an important feature of human governance. The capacity to propose a constitution and the authority to adopt one are not necessarily the same thing. An institutional entrepreneur may design a rule, but a community must still decide whether that rule is acceptable. NB04 represents this distinction computationally.

The vote also makes heterogeneity consequential. Agents do not need to value openness, verification, incentives, or flexibility in exactly the same way. A constitution becomes operational because it survives a collective selection process, not because the programmer labels it “best.”

Pedagogically, this is the moment when the system moves from discussion to commitment. Before the vote, constitutions are ideas. After the vote, one proposal becomes a rule-bearing object that constrains subsequent collaboration.

The adopted constitution therefore creates a form of endogenous path dependence. Later coalition structure is no longer determined only by evidence and complementarity. It is filtered through rules that the agents themselves selected. The society has begun to design part of the environment in which its future actions will occur.

Collective choice is kept deterministic. Each socially aware agent evaluates the constitutions through a transparent utility function reflecting its specialty-specific preference for openness, together with textual evidence of verification, autonomy, and diversity protections. A small seeded idiosyncratic term prevents perfectly mechanical bloc voting without making the experiment irreproducible.

This design choice is pedagogically important. The LLM created the alternatives, but it does not choose its own preferred constitution. Institutional legitimacy comes from a separate, inspectable choice mechanism. Ties are resolved transparently using mean utility and then a stable identifier. For students of finance, the analogy is familiar: product design, model generation, governance review, and authorization should not collapse into one opaque act. Separation of functions is itself a governance technology.

Code Unit 8: The Adopted Constitution Governs Sharing and Coalition Formation

This unit tests whether constitutional choice has real consequences. A constitution is meaningful only if it changes behavior. The adopted social compact is therefore translated into operational thresholds and rules that govern evidence sharing, complementary responses, coalition membership, and related collaboration decisions.

The conceptual sequence now becomes recursive. Earlier discoveries created the need for collaboration. Collaboration created the need for rules. The selected rules now feed back into the architecture of collaboration. Institutional design becomes an input into network formation.

For example, a constitution with a high sharing threshold may preserve confidentiality but reduce the amount of information available to potential collaborators. A more open constitution may accelerate collective synthesis while increasing the risk of herding or information leakage. Verification requirements may slow coalition growth but improve evidentiary quality. Reward rules can affect which agents find participation attractive. Merge and split rules influence whether the network concentrates around one hypothesis or preserves multiple competing structures.

The code operationalizes these ideas in a deliberately simplified manner. This is not intended as a complete theory of constitutional economics for artificial agents. The purpose is to make one causal idea visible: rules alter the graph.

That insight distinguishes NB04 from a conventional multi-agent workflow. In a fixed workflow, architecture is a parameter chosen by the designer. Here architecture is partly an outcome of an institutional process that occurred inside the system.

The unit also prepares the final graph by linking agents through complementarities under the constraints of the adopted constitution. Coalition formation is therefore neither purely spontaneous nor centrally assigned. It is spontaneous within a self-selected rule system.

This is a more realistic conception of organization. Human institutions rarely consist of unconstrained individuals connecting arbitrarily. Relationships form inside rules, norms, incentives, and permissions. NB04 begins to give artificial coalitions the same layered structure.

A constitution is meaningful only if it changes behavior. The selected sharing threshold determines which discoveries become public signals. The joining, verification, reward, and anti-herding principles shape which agents can form relationships around those signals. The constitution therefore ceases to be text and becomes executable institutional logic.

This is the notebook’s central causal claim. The final coalition network is not generated only by the acquisition

problem. It is generated by the interaction between the problem and the rules the agents selected. Change the constitution and one should expect a different pattern of disclosure, participation, verification, and coalition formation. Architecture has therefore become partly endogenous at two levels: relationships emerge, and the rules influencing those relationships have also been generated and selected within the system.

Code Unit 9: Emergent Graph and Collective Hypothesis

After search, semantic valuation, pheromones, social discovery, public communication, constitutional design, voting, and governed sharing, the system can finally be viewed as a network. This unit constructs the emergent collaboration graph and computes the collective hypothesis produced by the swarm.

The graph is important because it is the visible organizational artifact of the preceding process. Nodes represent agents. Edges represent relationships that became meaningful under the adopted constitution. The network was not drawn first and populated later. It is the accumulated consequence of discoveries, competence gaps, social awareness, institutional proposals, collective choice, and constitutional constraints.

The collective score similarly combines more than raw evidence. It incorporates LLM-valued discoveries, diversity, and social activation. The target hypothesis is therefore produced by a system whose epistemic and institutional states have co-evolved.

This is the point at which the phrase “the problem helps produce the organization that solves the problem” becomes concrete. The acquisition mandate created distributed uncertainty. Distributed uncertainty generated valuable local discoveries. Those discoveries exposed missing expertise. Environmental traces revealed other minds. Other minds made communication useful. Communication created a constitutional problem. Constitutional choice shaped coalition formation. The resulting coalition then produced the collective hypothesis.

The causal direction is therefore not simply organization-to-solution. It is partially circular:

Pedagogically, students should inspect the graph not merely as a visualization but as a historical record. Every edge has an institutional genealogy. It exists because earlier stages made that relationship possible and legitimate. This is what makes the network endogenous in a richer sense than a dynamically routed workflow.

Once relationships exist, the swarm can be represented as a graph. Agents are nodes; coalition relationships are edges; repeated interactions strengthen connections. The graph is not a decorative visualization. It is the institutional footprint of the experiment: a record of who found whom useful under the adopted rules.

The collective hypothesis then aggregates several dimensions of evidence: total LLM-valued discovery strength, average relevance, specialty diversity, public signaling, and coalition activity. The purpose is not to claim that this particular score is an optimal acquisition model. It is a pedagogical device showing how local evidence and social organization can jointly influence a final decision. Success means that the collective process recovers the hidden benchmark C0173. More important, however, is the path by which the answer was produced.

Code Unit 10: Visualization, Diagnostics, and the Institutional Learning Summary

The final code unit makes the experiment inspectable. It visualizes the constitutional swarm, prints the major state transitions, and summarizes what changed between the beginning and the end of the run. Visualization is not an ornamental final step. In emergent systems, observability is part of the architecture. If relationships, coalitions, and rules can arise dynamically, a human operator needs tools for understanding what the system became.

The diagnostic outputs should be read at several levels. At the epistemic level, which discoveries were judged valuable and why? At the social level, which agents became aware of others and which messages activated collaboration? At the constitutional level, what alternatives were proposed, how were they evaluated, and which compact won? At the organizational level, what coalition graph emerged under the selected rules? At the task level, did the collective hypothesis identify the intended acquisition target?

The final summary also clarifies the notebook’s relationship to NB03. The earlier swarm made coalition topology endogenous while leaving most collaboration rules exogenous. NB04 adds a constitutional layer. Agents begin isolated, use LLM judgment to interpret discoveries, infer the existence of other minds, activate a public communication mechanism, generate competing institutional designs, select one collectively, and allow that choice to shape the final network.

The experiment should not be interpreted as a claim that artificial societies have been fully achieved. Important substrate remains designed. Instead, NB04 demonstrates a direction of travel: move selected organizational decisions

from the programmer into the system while preserving explicit boundaries and auditability.

The final object is therefore not merely a better acquisition recommendation. It is an example of institutional learning. The system has produced a solution and, at the same time, has created a rule-bearing organizational structure through which that solution became possible.

The final unit turns the experiment into an inspectable object. One visualization shows the social architecture that formed; another shows the ranking of candidate companies after constitutional formation. Diagnostics connect the final outcome back to the institutional history: how many agents participated, how many relationships emerged, which constitution was adopted, what candidate won, and how much LLM capacity was consumed.

This final act of inspection is not cosmetic. As systems become more endogenous, governance must move beyond asking why a single model produced a particular answer. Practitioners must be able to ask how a network formed, which rules governed it, where information moved, and how institutional choices influenced the decision. Interpretability therefore expands from model interpretability to organizational interpretability.

What Actually Becomes Endogenous?

The strongest way to understand the notebook is to separate several layers of endogeneity. Discovery is endogenous because different specialists independently determine which evidence becomes salient. Relationship formation is endogenous because useful connections arise from traces, public signals, and complementary expertise rather than from a fixed organizational chart. Coalition structure is endogenous because the final graph is produced by local interactions. Most importantly, part of the institutional logic becomes endogenous because agents generate alternative constitutions and collectively select the compact that governs subsequent sharing and coalition formation.

But the substrate remains designed. The agents do not invent Python, the Anthropic API, their identifiers, the synthetic documents, the existence of persistence, or the voting machinery. Nor do they have unrestricted authority to rewrite the environment. The correct claim is therefore not that the system is fully autonomous. The correct claim is that selected design decisions have migrated inward.

This distinction is especially important in finance, where the word autonomy can obscure accountability. A useful autonomous system is not one in which every boundary disappears. It is one in which the system can adapt inside explicit boundaries and can explain enough of its organizational history for humans to audit what happened. The Constitutional Swarm therefore suggests a governance-first definition of advanced agentic architecture: greater internal freedom should be accompanied by stronger provenance, permissions, resource constraints, and organizational observability. [4, 5]

The progression can be summarized conceptually as endogenous discovery, endogenous coalition, endogenous constitution, and finally endogenous institution. The last term should be used carefully. The notebook creates only a primitive institution, not a full artificial society. Yet it demonstrates the essential mechanism: rules are no longer merely documentation surrounding the agents. They become operational artifacts that influence behavior after being generated and selected within the system.

Why the Hybrid Architecture Matters

A tempting reaction to the rise of frontier models is to place an LLM everywhere. The notebook takes the opposite position. It uses deterministic logic wherever the rule is already known and reserves language-model judgment for places where semantics, analogy, interpretation, or institutional invention are genuinely open-ended. This hybrid design is one of the paper's most practical lessons.

Private search can be bounded and reproducible. Candidate compression can be deterministic. Pheromone thresholds can be explicit. Voting can be transparent. Graph construction and scoring can be inspected. Claude is introduced where a fixed rule would be brittle: deciding whether a clue materially advances the mandate, explaining why it matters, identifying missing expertise, compressing an insight into a socially useful trace, and proposing coherent constitutions under novel circumstances.

For financial institutions, this division of labor offers a path between two unsatisfactory extremes. At one extreme lies the deterministic fortress: highly controlled but often unable to interpret ambiguous, unstructured situations. At the other lies the unconstrained AI agent: flexible but difficult to audit and authorize. The Constitutional Swarm demonstrates a middle architecture in which probabilistic intelligence is embedded inside deterministic scaffolding. The scaffolding does not eliminate emergence; it makes emergence legible.

This is also why the notebook should not be read as a benchmark proving that a constitutional swarm outperforms conventional M&A analysis. Its purpose is architectural. The synthetic acquisition case provides a controlled

laboratory in which the causal sequence can be observed. The educational outcome is the ability to distinguish model intelligence, agent intelligence, collective intelligence, and institutional intelligence rather than collapsing all four into the phrase “AI system.”

Governance: From Model Interpretability to Organizational Interpretability

Conventional model governance asks questions such as: What data entered the model? Which version produced the output? Why did the model make this prediction? Who approved deployment? Those questions remain necessary, but they are no longer sufficient when the system can create relationships and choose rules.

A constitutional swarm requires additional levels of observability. At the agent level, practitioners need provenance for claims, tool calls, and evidence. At the relationship level, they need to know who communicated with whom and what triggered the connection. At the coalition level, they need membership, purpose, resources, and decision rights. At the constitutional level, they need the rules that were proposed, the mechanism by which one was adopted, and the conditions under which it can change. At the ecosystem level, they need to observe concentration, dependency, information flow, resource consumption, and emergent authority.

This produces a governance paradox. The value of endogenous architecture comes from allowing useful relationships that the designer did not specify in advance. Requiring every future relationship to be predetermined would destroy the very capability being studied. Governance must therefore shift from exhaustive control toward constitutional constraint. We may not know which coalition will form, but we can specify what information may never be shared. We may not know which social compact will be proposed, but we can require auditability, reversibility, and bounded permissions. We may not prescribe every reasoning path, but we can require provenance and resource limits.

For finance, this is not an abstract concern. Agentic systems may eventually span research, valuation, risk, compliance, execution, and client communication. If such systems are allowed to adapt their internal organization, governance must become capable of inspecting organizational state, not merely model state. The question changes from “why did the model say this?” to “how did this artificial organization come to act this way?”

From Constitutional Swarms to Artificial Societies

The notebook ends at a boundary rather than a destination. All agents still share one acquisition mandate. That common objective is a hidden source of order. A stronger experiment would remove it. Imagine unrelated agents working on unrelated tasks, sharing only a minimal public communication surface. Most of the time they would remain independent. Occasionally one would reach a competence frontier, publish an appeal, and attract an unexpected capability from another domain. Temporary coalitions could form, borrow or invent constitutions, solve a problem, dissolve, and leave behind reusable institutional memory.

This is the direction suggested by the general framework in *The Bulletin Board*. The deepest architecture may not be a giant predefined hierarchy of agents. It may be a minimal environment in which intelligent actors can discover that they need one another. Repeated interaction could then create reputation, reusable tools, social conventions, and a library of institutional forms. Knowledge would transfer not only through retraining but through organizational reuse. A constitution that worked for one coalition could become a cognitive artifact available to another. [3, 2]

Such a system would be more powerful and more dangerous. Cross-domain collaboration could produce creative solutions that no designer anticipated, but the same openness could produce unexpected information flows, concentrations of authority, or unstable conventions. The appropriate objective is therefore neither complete central control nor unrestricted emergence. It is bounded emergence: environments open enough to permit unexpected collaboration but constrained enough that novelty remains observable, reversible, and governable.

The Constitutional Swarm is a small step toward that larger research program. Its importance lies not in the number of agents or the sophistication of the acquisition case. It lies in the relocation of the design question. Instead of asking only how to build a smarter model, we begin asking how to build an environment in which intelligent components can discover relationships, negotiate institutions, and create useful organizations without surrendering human accountability.

Conclusion: The Problem Begins to Design the Organization

The Constitutional Swarm begins with a familiar financial mandate and ends with a different object of study. Eighteen specialized agents search a synthetic acquisition universe. Claude helps interpret selected discoveries rather than replacing the search process. Valuable discoveries leave traces. Traces reveal other minds. Other minds make public communication meaningful. Communication exposes the need for governance. Agents propose constitutions. The population selects one. The adopted compact changes what is shared and how coalitions form. The final recommendation therefore emerges from both evidence and institution.

The conceptual progression is more important than the specific acquisition result. In a traditional model, the organization is outside the model. In a conventional multi-agent workflow, intelligence is distributed but the workflow is still designed. In the Constitutional Swarm, selected elements of organization become outputs of the intelligent process itself. The system begins to learn not only about the world but about how it should organize cognition in order to understand the world.

This is why the experiment belongs in a course for financial practitioners. The future of AI in finance is unlikely to be defined only by better forecasting models or larger chat interfaces. It will increasingly involve compound systems in which deterministic models, language models, tools, memory, agents, permissions, and governance mechanisms interact. Understanding those systems requires a new literacy: practitioners must be able to reason about architectures, incentives, information flows, institutional rules, and emergent organization.

The notebook does not argue for abandoning hierarchy or human governance. It shows why those mechanisms must evolve. As more organizational decisions become adaptive, governance must become constitutional: clear about boundaries, explicit about permissions, rigorous about provenance, and capable of observing structures that were not specified in advance.

The central lesson can therefore be stated simply. The problem does not merely select the answer. The problem can help produce the organization—and eventually some of the rules of the organization—that produces the answer. That is the step from endogenous coalition to endogenous constitution, and from a collection of artificial agents toward the first primitive forms of an artificial institution.

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